edu.mit.rocket.team.amphyrus
instrument
internal
telemetry
util
RTMag
extends RTPeripheral
-magX: float
-magY: float
-magZ: float
+setup()
+getMagX(): float
+getMagY(): float
+getMagZ(): float
+backdoorFudge(fudged: RTFudgedData)
RTGyro
extends RTPeripheral
-gyroX: float
-gyroY: float
-gyroZ: float
+setup()
-begin()
-configure()
+getGyroX(): float
+getGyroY(): float
+getGyroZ(): float
+backdoorFudge(fudged: RTFudgedData)
RTGPS
extends RTPeripheral
-latitude: float
-longitude: float
-altitude: float
-PDOP: float
-EDOP: float
-HDOP: float
-hasFix: boolean
-gpsChar: char[]
-rawGPS: float
-gpsAltOffset: float
-newGpsAlt: float
-gpsSer: RTHardwareSerial
+setup()
+updateAndParse()
+getLatitude(): float
+getLongitude(): float
+getAltitude(): float
+getPDOP(): float
+getVDOP(): float
+getHDOP(): float
+getFix(): boolean
+backdoorFudge(fudged: RTFudgedData)
-gpsUpdateData(length: int): boolean
-gpsParse(data: char[], length: int)
-convertToDecimalPlaces(): float
-indexOf(arr: char[], target: char, start: int): int
RTBaro
extends RTPeripheral
-rawPressure: float
-rawTemperature: float
-temperature: float
-altitude: float
-pressure: float
-movingAvg: float
+setup()
-setartPressureConversion()
+startTemperatureConversion()
+getRawPressure(): float
+getRawTemperature(): float
+pressure(): float
+temperature(): float
+altitude(): float
+altitudeAvg(): float
+backdoorFudge(fudged: RTFudgedData)
RTAccel
extends RTPeripheral
-accelX: float
-accelY: float
-accelZ: float
+setup()
+update()
+getAccelX(): float
+getAccelY(): float
+getAccelZ(): float
+backdoorFudge(fudged: RTFudgedData)
RTPowerBoard
RTFlashDriver
RTTelemetryEngine
RTRangingEngine
hardware
RTHardwareSerial
RTInstrument
+setup()
+backdoorFudge(fudged: RTFudgedData)
RTController
data
RTMagData
extends RTPeripheralData
-magX: float
-magY: float
-magZ: float
++constructor(x: float, y: float, z: float)
+getMagX(): float
+getMagY(): float
+getMagZ(): float
+setMagX(magX: float)
+setMagY(magY: float)
+setMagZ(magZ: float)
RTGyroData
extends RTPeripheralData
-gyroX: float
-gyroY: float
-gyroZ: float
++constructor(x: float, y: float, z: float)
+getGyroX(): float
+getGyroY(): float
+getGyroZ(): float
+setGyroX(gyroX: float)
+setGyroY(gyroY: float)
+setGyroZ(gyroZ: float)
RTAccelData
extends RTPeripheralData
-accelX: float
-accelY: float
-accelZ: float
++constructor(x: float, y: float, z: float)
+getAccelX(): float
+getAccelY(): float
+getAccelZ(): float
+setAccelX(accelX: float)
+setAccelY(accelY: float)
+setAccelZ(accelZ: float)
RTBaroData
extends RTPeripheralData
-rawPressure: float
-rawTemperature: float
-temperature: float
-altitude: float
-pressure: float
++constructor(rP: float, rT: float, t: float, a: float, p: float)
+getRawPressure(): float
+getRawTemperature(): float
+getTemperature(): float
+getAltitude(): float
+getPressure(): float
+setRawPressure(rawPressure: float)
+setRawTemperature(rawTemperature: float)
+setTemperature(temperature: float)
+setAltitude(altitude: float)
+setPressure(pressure: float)
RTGPSData
extends RTPeripheralData
-latitude: float
-longitude: float
-altitude: float
-PDOP: float
-EDOP: float
-HDOP: float
-hasFix: boolean
++constructor(lat: float, lon: float, alt: float, p: float, v: float, h: float)
+getLatitude(): float
+getLongitude(): float
+getAltitude(): float
+getPDOP(): float
+getVDOP(): float
+getHDOP(): float
+getHasFix(): float
+setLatitude(lat: float)
+setLongitude(lon: float)
+setAltitude(alt: float)
+setPDOP(p: float)
+setVDOP(v: float)
+setHDOP(h: float)
+setHasFix(hasFix: boolean)
RTFudgedData
info.openrocket.core.simulation.listeners
FlightComputerSimulationListener
initialStatus: SimulationStatus
latestStatus: SimulationStatus
lastStat: SimulationStatus
thsFinalToNotify: TabControlledPropsandFinSet
recentAltMA: double
TIME_DELAY_MOTOR: double
amplitude_randomness_size: double
ABORT_AT_APOGEE: boolean
loopStart: boolean
servoOnpCount: double
lastServoCommandTimestamp: double
flagFirstDebugFlag: boolean
pastOmegaZ: ArrayList<Double>
pastThetaZ: ArrayList<Double>
finTabAngLeftLog: ArrayList<Double>
rightTabAngLog: ArrayList<Double>
rvelAlLog: ArrayList<Double>
ClrLog: ArrayList<Double>
QLog: ArrayList<Double>
ClldrstALog: ArrayList<Double>
altitudeMeasuredList: ArrayList<Double>
SERVO_REPRESS_TIME: double
servoRangeAngle: double (final)
roundFinRecursth: boolean
+constructor()
// Simulation Progression
+startSimulation(status: SimulationStatus) [callback from SimulationListenerHelper] calls PC.init()
+preStep(status: SimulationStatus) [callback from SimulationListenerHelper]
+postStep(status: SimulationStatus) [callback from SimulationListenerHelper] calls PC.loop()
// Fin Tab Actuation Wrapper
+getThsFinsToNotifyTabs(status: SimulationStatus): TabControlledPropsandFinSet
+setFinTabAngle(newAngle: double)
+setFinTabAngleLeftLog(newAngle: double)
+getFinTabAngleLeftLog(): double
// Utilities
+toEulerAngle(q: Quaternion): Coordinate